

Notes: Coval & Stafford (JFE, 2007)

Asset Fire Sales (and Purchases) in Equity Markets

Contents

1	Big picture	3
2	Incremental contribution	3
3	Data and measurement	3
3.1	Mutual fund flows	3
3.2	Fund-level behavior	3
3.3	Stock-level pressure	3
4	Empirical specifications	4
4.1	Expected flows	4
4.2	Event-time abnormal returns	4
4.3	Calendar-time portfolio regressions	4
5	Identification and empirical design	4
6	Section-by-section study guide	4
7	Tables and figures	5
7.1	Table 1: Regressions explaining mutual fund flows (1980–2004)	5
7.2	Table 2 and Figure 1: Fund behavior and flow-driven adjustments	5
7.3	Table 3: Characteristics of stocks involved in flow-motivated trading	6
7.4	Table 4: Monthly cumulative average abnormal returns around mutual fund transactions	7
7.5	Figures 2 and 3: Fire-sale and voluntary-sale CAARs	8
7.6	Table 5: Alternative pressure measures	8
7.7	Table 6: Mutual fund transaction characteristics by capital flow decile	8
7.8	Table 7: Calendar-time portfolio regressions of recent flow-induced transactions	9
7.9	Table 8: Calendar-time portfolio regressions of expected flow-induced transactions	10
7.10	Table 9: Return predictability of anticipated pressure	10
8	Detailed result map	11
9	Short summary	11
10	Conclusion	11
10.1	Core conclusion	11
10.2	Incremental contribution	12
10.3	Economic intuition	12

10.4 Implications	12
10.5 Limitations	12
10.6 Takeaway	12

1 Big picture

- **Question.** Do open-end mutual fund flows create asset fire sales and inflow-driven purchases in public equity markets?
- **Setting.** The paper uses mutual fund holdings, flows, and stock returns from 1980 to 2004.
- **Mechanism.** Funds with severe outflows must reduce existing positions. Funds with severe inflows often expand existing positions. When many funds overlap in holdings, their trades create stock-level price pressure.
- **Main result.** Fire-sale stocks fall sharply around the event quarter and rebound later. Inflow-driven purchase stocks display the mirror form of positive pressure and later reversal.
- **Economic takeaway.** Even highly liquid equity markets can experience multi-month price pressure when institutional investors are simultaneously constrained.

2 Incremental contribution

- The paper moves the fire-sale mechanism from opaque corporate or real-asset settings into transparent public equity markets.
- Mutual fund flows provide a clean financing shock that helps separate financial distress from stock-specific information.
- The paper documents symmetric institutional pressure: forced sales and inflow-driven purchases.
- It shows persistence over months rather than days and links the mispricing to feasible liquidity-provision and front-running strategies.
- It connects organizational design of money management to asset-pricing distortions.

3 Data and measurement

3.1 Mutual fund flows

$$Flow_{j,t} = \frac{TNA_{j,t} - TNA_{j,t-1}(1 + Return_{j,t})}{TNA_{j,t-1}}.$$

Detailed interpretation: This removes capital gains and losses from changes in assets under management and leaves investor capital flow.

3.2 Fund-level behavior

For each fund-quarter, the authors calculate the fraction of positions maintained, expanded, reduced, or eliminated, and the average change in holdings. Funds are sorted by actual and expected flows.

Detailed interpretation: This step verifies that flows translate into trading in current holdings.

3.3 Stock-level pressure

$$PRESSURE_{1,i,t} = \frac{\sum_j \max(0, \Delta Holdings_{j,i,t}) 1\{Flow_j > 90th\} - \sum_j \max(0, -\Delta Holdings_{j,i,t}) 1\{Flow_j < 10th\}}{AvgVolume_{i,t-12:t-6}}.$$

$$PRESSURE2_{i,t} = \frac{\sum_j \max(0, Flow_{j,t}) \max(0, \Delta Holdings_{j,i,t}) - \sum_j \max(0, -Flow_{j,t}) \max(0, -\Delta Holdings_{j,i,t})}{AvgVolume_{i,t-12:t-6}}.$$

$PRESSURE3_{i,t}$ = same numerator as $PRESSURE1$ scaled by shares outstanding.

Detailed interpretation: Negative pressure identifies fire-sale candidates; positive pressure identifies inflow-driven purchases.

4 Empirical specifications

4.1 Expected flows

$$Flow_{j,t} = \alpha_t + \sum_k \beta_k Flow_{j,t-k} + \sum_k \gamma_k Return_{j,t-k} + \varepsilon_{j,t}.$$

4.2 Event-time abnormal returns

$$AAR_t = \frac{1}{N_t} \sum_i (R_{i,t} - R_t^{MF \text{ universe}}), \quad CAAR_{\tau_1, \tau_2} = \sum_{t=\tau_1}^{\tau_2} AAR_t.$$

4.3 Calendar-time portfolio regressions

$$R_{p,t+1} - R_{f,t+1} = \alpha + \beta(R_{m,t+1} - R_{f,t+1}) + sSMB_{t+1} + hHML_{t+1} + uUMD_{t+1} + \varepsilon_{t+1}.$$

Detailed interpretation: The intercept is the abnormal return to trading against flow-induced pressure.

5 Identification and empirical design

The core challenge is separating liquidity-motivated trading from information-motivated trading. The paper addresses this by using severe mutual fund flows as financial pressure, requiring overlapping institutional ownership, testing for post-event reversals, comparing forced and voluntary sales, and checking robustness across pressure measures. The main limitation is that holdings are quarterly and some trades may mix information and liquidity motives.

6 Section-by-section study guide

- **Introduction:** motivates fire sales in a market where they should be least likely.
- **Section 2:** documents predictable flows and fund behavior under flow pressure.
- **Section 3:** presents event-time price-pressure evidence.
- **Section 4:** shows robustness and studies trade-selection behavior.
- **Section 5:** shows liquidity-provision and front-running returns.
- **Section 6:** discusses persistence, economic significance, and implications.

7 Tables and figures

7.1 Table 1: Regressions explaining mutual fund flows (1980–2004)

Read: Lagged flows and lagged returns predict future flows, justifying expected-flow measures.

Detailed interpretation: This table makes the front-running analysis possible.

Economic message. Fund flows are persistent and forecastable.

Table 1

Regressions explaining mutual fund flows (1980–2004)

This table reports results from regressions of mutual fund flows on lagged fund flows and lagged fund returns. Mutual fund flows are measured as a percentage of beginning-of-period total net assets (TNA). Mutual fund flows are estimated as the percentage change in TNA over the quarter controlling for capital gains and losses of the initial holdings: $[TNA_t - TNA_{t-1} \times (1 + Return_t)] / TNA_{t-1}$. Quarterly regressions use quarterly observations on flows and returns, while monthly regressions use monthly observations of flows and returns. Fama-MacBeth regression coefficients are the time-series average of periodic cross-sectional regression coefficients, with t -statistics calculated using the time-series standard error of the mean. The reported R^2 is the average across all cross-sectional regressions. The pooled regression results are based on OLS coefficients, where the mean of each variable has been subtracted. The number of observations is denoted by N , and t -statistics are in parentheses.

	Quarterly		Monthly	
	Fama-MacBeth	Pooled	Fama-MacBeth	Pooled
Intercept	−0.0211 (−5.61)	0.0000 (0.00)	−0.0069 (−7.25)	0.0000 (0.00)
flow($t-1$)	0.2499 (9.35)	0.2887 (40.22)	0.1329 (9.27)	0.1673 (37.31)
flow($t-2$)	0.1762 (9.55)	0.1494 (20.52)	0.1684 (15.95)	0.1672 (36.97)
flow($t-3$)	0.1264 (6.50)	0.0979 (13.39)	0.1617 (14.33)	0.1322 (28.94)
flow($t-4$)	0.0852 (5.15)	0.0584 (8.17)	0.0711 (6.54)	0.0605 (13.23)
flow($t-5$)	0.0032 (0.19)	0.0311 (4.38)	0.0682 (6.77)	0.0714 (15.72)
flow($t-6$)	−0.0079 (−0.58)	−0.0089 (−1.31)	0.0473 (4.64)	0.0346 (7.69)
flow($t-7$)	−0.0218 (−1.48)	0.0032 (0.49)	0.0243 (2.55)	0.0258 (5.79)
flow($t-8$)	0.0659 (5.15)	0.0513 (8.79)	0.0276 (2.98)	0.0196 (4.47)
flow($t-9$)			0.0391 (4.47)	0.0352 (8.12)
flow($t-10$)			0.0150 (1.72)	0.0104 (2.45)
flow($t-11$)			0.0067 (0.80)	0.0136 (3.31)
flow($t-12$)			0.0179 (2.05)	0.0180 (4.60)
Return($t-1$)	0.2867 (12.82)	0.1909 (23.04)	0.1196 (12.97)	0.0736 (23.88)
Return($t-2$)	0.1795 (7.74)	0.0579 (7.08)	0.0845 (7.96)	0.0369 (12.13)
Return($t-3$)	0.1794 (7.21)	0.0760 (9.50)	0.0814 (7.43)	0.0334 (11.04)

(a) Table 1, first page

J. Carol E. Stafford / Journal of Financial Economics 86 (2007) 479–512 485

Table 1 (continued)

	Quarterly		Monthly	
	Fama-MacBeth	Pooled	Fama-MacBeth	Pooled
Return($t-4$)	0.0961 (3.52)	0.0383 (4.82)	0.0679 (6.56)	0.0241 (7.99)
Return($t-5$)	−0.0169 (−0.81)	0.0488 (6.36)	0.0621 (5.73)	0.0150 (4.98)
Return($t-6$)	−0.0313 (−1.22)	0.0036 (0.49)	0.0324 (3.51)	0.0132 (4.38)
Return($t-7$)	0.0552 (2.66)	0.0077 (1.04)	0.0504 (4.86)	0.0060 (2.01)
Return($t-8$)	0.0387 (1.53)	−0.0024 (−0.34)	0.0240 (2.20)	0.0037 (1.24)

(b) Table 1, continued

7.2 Table 2 and Figure 1: Fund behavior and flow-driven adjustments

Read: Extreme outflows lead funds to reduce or eliminate positions; extreme inflows lead funds to expand positions. Figure 1 shows the monotone relation between capital flow deciles and net adjustments.

Detailed interpretation: This is the behavioral foundation of the fire-sale mechanism.

Economic message. Investor flows become demand shocks for individual portfolio holdings.

Table 2

Mutual fund trading associated with actual and predicted flows (1980–2004)

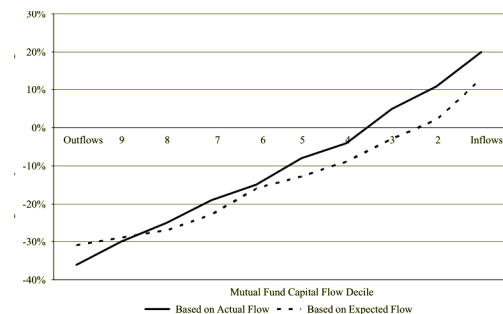
This table reports how quarterly mutual fund holdings change conditional on actual and expected fund flows. Mutual fund flows are measured as a percentage of beginning-of-period total net assets (TNA). Mutual fund flows are estimated as the percentage change in TNA over the quarter controlling for capital gains and losses of the initial holdings: $[TNA_t - TNA_{t-1} \times (1 + Return_t)] / TNA_{t-1}$. Expected flow is estimated via Fama-MacBeth regressions of quarterly flows on lagged flows and returns, where coefficients are the time-series average of periodic cross-sectional regression coefficients. Fund-quarter observations with available flow data ($N = 142,720$) are sorted according to actual quarterly fund flow (Panels A and B) and expected quarterly flow (Panel C). Within each flow decile, Panel A reports the fund flow, expected flow, number of holdings, past 12-month fund return, and cash holdings averaged across all funds in the decile. The final two columns of Panel A report for each flow decile the average fund's size decile of its holdings (using the NYSE break-points reported on Ken French's website) and the average volume decile. Panels B and C report for the average fund the fraction of positions that were maintained, expanded, reduced, or eliminated. The final column reports the average fund's percent change in positions held at the start of the quarter.

Decile	Flow (%)	E[Flow] (%)	Prior fund return (%)	Average number of holdings	Average Cash/TNA (%)	Average holding size decile	Average holding volume decile
<i>Panel A: Fund and holding characteristics (sorted on actual flows)</i>							
1 (Inflow)	23.7	6.9	18.7	124.3	5.7	4.4	2.3
2	8.1	4.3	14.9	143.8	5.5	3.9	2.0
3	3.6	2.3	12.9	167.1	5.7	3.6	1.9
4	1.0	0.6	9.9	160.4	5.2	3.5	1.8
5	-0.6	-0.6	8.1	143.6	5.1	3.3	1.8
6	-2.0	-1.6	6.0	134.3	4.7	3.4	1.8
7	-3.4	-2.6	3.3	130.8	4.3	3.3	1.7
8	-4.9	-3.4	1.4	130.4	3.7	3.1	1.6
9	-7.0	-4.4	0.7	134.1	3.3	3.4	1.7
10 (Outflow)	-12.5	-5.1	2.8	117.1	3.6	3.5	1.8

Decile	Flow (%)	Fraction of positions				Average change in holding (%)
		Maintained	Expanded	Reduced	Eliminated	
<i>Panel B: Fund trading behavior (sorted on actual flow)</i>						
1 (Inflow)	23.7	0.28	0.46	0.11	0.15	0.8%
2	8.1	0.36	0.38	0.14	0.13	-5.0%
3	3.6	0.41	0.32	0.15	0.12	-5.4%
4	1.0	0.41	0.27	0.18	0.13	-7.9%
5	-0.6	0.45	0.23	0.19	0.12	-8.2%
6	-2.0	0.42	0.21	0.23	0.13	-9.9%
7	-3.4	0.41	0.19	0.24	0.14	-13.2%
8	-4.9	0.36	0.20	0.31	0.14	-14.1%
9	-7.0	0.29	0.20	0.33	0.17	-16.8%
10 (Outflow)	-12.5	0.27	0.18	0.37	0.17	-20.5%

Decile	E[Flow] (%)	Fraction of positions				Average change in holding (%)
		Maintained	Expanded	Reduced	Eliminated	
<i>Panel C: Fund trading behavior (sorted on expected flow)</i>						
1 (Inflow)	9.6	0.35	0.39	0.13	0.13	-1.9%
2	3.9	0.41	0.31	0.17	0.12	-6.1%
3	1.7	0.43	0.27	0.18	0.12	-7.3%

(a) Table 2, first page



(b) Figure 1

Table 2 (continued)

Decile	E[Flow] (%)	Fraction of positions				Average change in holding (%)
		Maintained	Expanded	Reduced	Eliminated	
Panel C: Fund trading behavior (sorted on expected flow)						
4	0.2	0.43	0.25	0.22	0.12	-8.0%
5	-1.1	0.42	0.23	0.23	0.13	-10.6%
6	-2.3	0.40	0.23	0.25	0.14	-11.8%
7	-3.3	0.37	0.20	0.28	0.15	-15.0%
8	-4.5	0.32	0.21	0.31	0.17	-17.6%
9	-5.9	0.29	0.20	0.33	0.16	-17.3%
10 (Outflow)	-9.6	0.26	0.19	0.33	0.17	-20.1%

Table 2, continued

7.3 Table 3: Characteristics of stocks involved in flow-motivated trading

Read: Event stocks are relatively large and growth-oriented compared with the CRSP universe.

Detailed interpretation: Fire-sale price pressure is not only a microcap illiquidity phenomenon.

Economic message. Institutional price pressure affects relatively liquid equities.

J. Coval, E. Stafford / Journal of Financial Economics 86 (2007) 479–512

calculate *PRESSURE* 2:

$$\begin{aligned} & \frac{\sum_j (\max(0, flow_{j,i}) \cdot \max(0, \Delta Holdings_{j,i}) - \sum_j (\max(0, -flow_{j,i}) \cdot \max(0, -\Delta Holdings_{j,i})))}{Avg Volume_{i,j-12:r-6}} \end{aligned} \quad (5)$$

This variable has the properties that both the severity of the flows (amount of duress the fund is facing) and larger transaction sizes (relative to the average volume of the stock) will each create more “pressure.”

The third pressure variable scales the original variable by shares outstanding rather than average volume to make sure that the denominator is not driving the results. This modification leads to the variable *PRESSURE* 3:

$$PRESSURE_{3,t} = \frac{\sum_j (\max(0, \Delta Holdings_{j,t}) | flow_{j,t} > Percentile(90th)) - \sum_j (\max(0, -\Delta Holdings_{j,t}) | flow_{j,t} < Percentile(10th))}{SharesOutstanding_{t-1}} \quad (6)$$

As with *PRESSURE_1*, we use the 10th and 90th percentiles of these pressure measures to identify firms facing extreme flow-induced trading.

Table 3 displays the distribution of the sample firms according to size and book-to-market quintiles using NYSE breakpoints (see Fama and French, 1992, 1993). Both flow-motivated sales and purchases have size distributions that are tilted towards large firms relative to the population of CRSP firms. More than 60% of all common stocks listed in

Table 3
Characteristics of stocks Involved in flow-motivated trading by mutual funds

Distribution of sample firms by various characteristics. All quintiles are defined using annual NYSE breakpoints. Stocks with *PRESSURE*₁ below the 10th percentile are determined to be fire sale stocks. Stocks with *PRESSURE*₁ above the 90th percentile are determined to be flow-motivated purchase stocks. *PRESSURE*₁ is a stock-level variable calculated as the quantity of shares purchased by mutual funds experiencing significant inflows, minus the quantity of shares sold by mutual funds with significant outflows, scaled by the average volume of the stock, requiring at least ten owners.

Sample	Quintile 5 (low) (%)	Quintile 4 (%)	Quintile 3 (%)	Quintile 2 (%)	Quintile 1 (high) (%)
<i>A: Market capitalization</i>					
Universe	61.2	14.4	9.5	7.6	7.2
Forced sales	3.2	10.8	21.2	33.5	31.2
Forced buys	3.2	9.7	17.7	32.6	36.8

Table 3

7.4 Table 4: Monthly cumulative average abnormal returns around mutual fund transactions

Read: Forced-sale stocks fall sharply around the event quarter and rebound later. The paper reports severe distressed selling associated with about -7.9% abnormal performance and a post-event rebound of about 6.13% over the next year.

Detailed interpretation: Drop plus reversal is the key evidence of temporary price pressure.

Economic message. Liquidity providers are compensated for absorbing forced sales.

[illegible]

7.5 Figures 2 and 3: Fire-sale and voluntary-sale CAARs

Read: Figure 2 shows a large decline and reversal around forced fire sales. Figure 3 shows voluntary sales without the same reversal.

Detailed interpretation: The comparison separates liquidity pressure from information-motivated selling.

Economic message. Forced sales create temporary mispricing; ordinary sales are more likely to convey information.

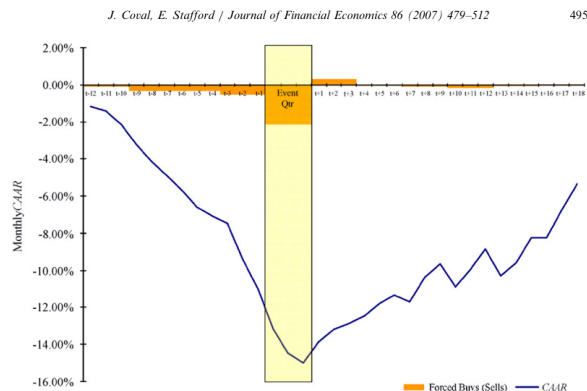


Fig. 2. Cumulative average abnormal returns around mutual fund fire sales. Cumulative average abnormal returns (CAARs) are measured monthly returns in excess of the equal-weighted average return of all stocks held by mutual funds at the start of the month. Each month the average of monthly abnormal return is calculated, and then the time-series mean and standard error of the mean are used for statistical inference. Transactions are identified as “forced” based on their capital flows as a percentage of their beginning-of-period total net assets. Fire sales are identified at the stock level based on net selling pressure below the 10th percentile.

$$\begin{aligned}
 \text{PRESSURE}_{1,t} &= \frac{\sum_j (\max(0, \Delta \text{Holdings}_{j,t}) | \text{flow}_{j,t} > \text{Percentile}(90\text{th}) - \sum_j (\max(0, -\Delta \text{Holdings}_{j,t}) | \text{flow}_{j,t} < \text{Percentile}(10\text{th}))}{\text{Avg Volume}_{t-12:t-6}}
 \end{aligned}$$

severe distressed selling of the same stock, the average abnormal stock return is -7.9% with a t -statistic of -3.45 . Over the quarter in which fire sales are occurring, the net forced selling pressure accounts for roughly 2% of average volume.

Importantly, the downward pattern in abnormal returns eventually reverses once the net forced sales dissipate. Over the 12 months that follow the fire sale quarter, average abnormal net forced selling pressure retreats to roughly 0%, and stock prices for the fire sale stocks rebound 6.13% over this period, with a t -statistic of 2.47. The modest statistical

(a) Figure 2

prices and, perhaps more commonly, mutual funds eliminating underperforming holdings Fig. 3.

In Table 5, we report a summary of the results using the samples identified with the alternative share-based pressure measures, along with the original results for easy comparison. As can be seen, the results for the fire sale sample are remarkably similar across the different pressure variables. This suggests that the results are not driven by abnormally low trading volume or by the cutoffs used to identify extreme outflows.

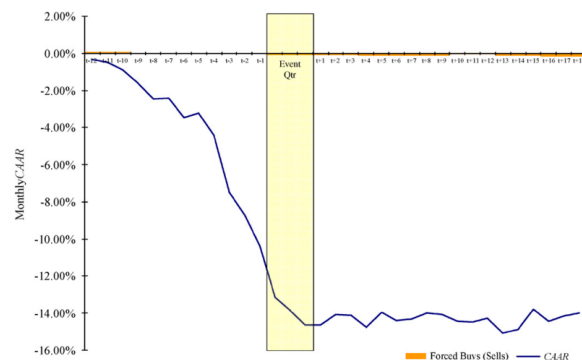


Fig. 3. Cumulative average abnormal returns around voluntary mutual fund sales. Cumulative average abnormal returns (CAARs) are measured monthly returns in excess of the equal-weighted average return of all stocks held by mutual funds at the start of the month. Each month the average of monthly abnormal return is calculated, and then the time-series mean and standard error of the mean are used for statistical inference. Transactions are not conditioned on the capital flows into the fund. Voluntary sales are identified at the stock level based on net selling pressure below the 10th percentile.

(b) Figure 3

7.6 Table 5: Alternative pressure measures

Read: Similar patterns appear across PRESSURE1, PRESSURE2, PRESSURE3, and unforced pressure comparisons.

Detailed interpretation: The results are robust to scaling and pressure definitions.

Economic message. Fire-sale evidence is not an artifact of one variable definition.

7.7 Table 6: Mutual fund transaction characteristics by capital flow decile

Read: Extreme-flow funds do not appear to avoid pressure costs by uniquely selecting only easy-to-trade holdings.

Table 5
Monthly cumulative average abnormal returns for stocks around flow-motivated trading by mutual funds identified with alternative measures
Cumulative average abnormal returns (*CAAR*) are measured monthly returns in excess of the equal-weighted average return of all stocks held by mutual funds at the start of the month. Flow-motivated buys and sales are identified as stocks with extreme values of one of three *PRESSURE* variables.

$$PRESSURE_{1,t} = \frac{\sum_{i=1}^N (\max(0, \Delta Holdings_{i,t}) I(\text{flow}_{i,t} > \text{Percentile}(90th)) - \sum_{i=1}^N (\max(0, -\Delta Holdings_{i,t}) I(\text{flow}_{i,t} < \text{Percentile}(10th)))}{\text{Avg Volume}_{i,t-12:t-4}}$$

$$PRESSURE_{2,t} = \frac{\sum_{i=1}^N (\max(0, \text{flow}_{i,t}) - \max(0, \Delta Holdings_{i,t})) - \sum_{i=1}^N (\max(0, -\text{flow}_{i,t}) - \max(0, -\Delta Holdings_{i,t}))}{\text{Avg Volume}_{i,t-12:t-4}}$$

$$PRESSURE_{3,t} = \frac{\sum_{i=1}^N (\max(0, \Delta Holdings_{i,t}) I(\text{flow}_{i,t} > \text{Percentile}(90th)) - \sum_{i=1}^N (\max(0, -\Delta Holdings_{i,t}) I(\text{flow}_{i,t} < \text{Percentile}(10th))))}{\text{Shares Outstanding}_{i,t-1}}$$

$$\text{Unforced_PRESSURE}_{i,t} = \frac{\sum_{i=1}^N \Delta Holdings_{i,t}}{\text{Avg Volume}_{i,t-12:t-4}}$$

J. Cough, E. Saffajou / Journal of Financial Economics 86 (2007) 417–432

407

*PRESSURE*₁ corresponds to the measure defined in Table 4. Flow-motivated transactions are identified at the stock level based on net buying pressure above the 90th percentile for buys and below the 10th percentile for sales. All measures require at least ten owners. All reported statistics are calculated from the time series of monthly averages of abnormal returns, requiring at least 25 events during the month. Test statistics are calculated using the standard error of the mean, and are in parentheses.

	<i>PRESSURE</i> ₁		<i>PRESSURE</i> ₂		<i>PRESSURE</i> ₃		<i>Unforced</i>	
	<i>CAAR</i> (%)	<i>t</i> -statistic	<i>CAAR</i> (%)	<i>t</i> -statistic	<i>CAAR</i>	<i>t</i> -statistic	<i>CAAR</i> (%)	<i>t</i> -statistic (%)
<i>Flow-motivated sales</i>								
Event Period [−6, 6]	−10.07	(−3.68)	−13.25	(−5.22)	−10.69	(−3.35)	−12.21	(−7.48)
Event Period [−3, 3]	−7.90	(−3.45)	−9.77	(−4.64)	−9.72	(−3.66)	−10.30	(−7.69)
Post Event [−1, 1, +3]	2.11	(1.77)	3.28	(2.65)	5.70	(2.85)	0.53	(0.50)
Post Event [−1, 1, +12]	6.13	(2.47)	5.76	(2.38)	8.10	(1.94)	0.37	(0.11)
Post Event [−1, 1, +18]	9.65	(3.47)	7.14	(2.39)	12.40	(2.44)	0.64	(0.22)
<i>Flow-motivated buys</i>								
Event Period [−6, 6]	6.10	(3.91)	12.59	(6.62)	11.68	(3.58)	10.64	(8.79)
Event Period [−3, 3]	3.64	(2.58)	8.83	(5.07)	6.42	(2.11)	8.25	(8.08)
Post Event [−1, 1, +3]	0.02	(0.01)	0.28	(0.07)	−2.33	(−1.50)	−0.01	(0.90)
Post Event [−1, 1, +12]	−0.73	(−0.21)	−1.95	(−1.04)	−6.26	(−2.52)	−0.49	(0.41)
Post Event [−1, 1, +18]	−3.86	(−1.87)	−5.46	(−2.21)	−9.53	(−3.00)	−2.80	(−0.99)

J. Cough, E. Saffajou / Journal of Financial Economics 86 (2007) 417–432

Detailed interpretation: The constraint is real: funds cannot fully neutralize redemption pressure through painless trade selection.

Economic message. Flow pressure transmits into portfolio holdings.

J. Cough, E. Saffajou / Journal of Financial Economics 86 (2007) 417–432

Table 6
Mutual fund trading response to actual flows (1980–2004)
This table reports how quarterly mutual fund holdings change conditional on actual fund flows. Mutual fund flows are measured as a percentage of beginning-of-period total net assets (*TNA*). Mutual fund flows are estimated as the percentage change in *TNA* over the quarter controlling for capital gains and losses of the initial holdings: $[TNA_t - TNA_{t-1} \times (1 + \text{Return}_t)] / TNA_{t-1}$. Fund-quarter observations with available flow data ($N = 142,720$) are sorted according to actual quarterly fund flow. Within each decile, Panel A reports the average relative size of fund holdings maintained, expanded, reduced, or eliminated. A holding's relative size is measured as the ratio of its market capitalization to the average market capitalization of the fund's current positions. Panel B reports the average relative trading volume of fund holdings maintained, expanded, reduced, or eliminated, with relative trading volume measured likewise. Panel C reports the average relative return of fund holdings maintained, expanded, reduced, or eliminated. Relative return is measured as the difference between the holding's return over the past year and the average fund holding's return over the past year.

Decile	Maintained (%)	Expanded (%)	Reduced (%)	Eliminated (%)
<i>Panel A: Relative size of holdings</i>				
1 (Inflow)	0.88	1.09	1.06	0.86
2	0.92	1.10	1.16	0.78
3	0.92	1.11	1.13	0.77
4	0.96	1.14	1.11	0.68
5	0.94	1.11	1.14	0.70
6	0.95	1.09	1.12	0.71
7	0.96	1.09	1.18	0.68
8	0.94	1.09	1.18	0.69
9	0.90	1.08	1.15	0.69
10 (Outflow)	0.91	1.03	1.18	0.78
<i>Panel B: Relative trading volume of holdings</i>				
1 (Inflow)	0.90	1.11	1.21	0.80
2	0.86	1.13	1.19	0.85
3	0.96	1.19	1.18	0.82
4	0.91	1.16	1.18	0.79
5	0.91	1.17	1.19	0.74
6	0.92	1.14	1.24	0.75
7	0.92	1.19	1.17	0.75
8	0.96	1.16	1.21	0.76
9	0.90	1.18	1.18	0.80
10 (Outflow)	0.92	1.21	1.17	0.81
<i>Panel C: Relative past return of holdings</i>				
1 (Inflow)	−2.9	1.1	9.5	−9.9
2	−3.6	0.8	8.9	−7.6
3	−3.0	1.8	5.7	−7.9
4	−2.8	1.1	5.2	−6.8
5	−2.8	2.6	4.8	−8.3
6	−2.1	2.0	4.3	−6.3
7	−1.7	1.5	2.6	−7.2
8	−2.5	2.8	1.4	−7.6
9	−1.4	3.7	1.9	−6.3
10 (Outflow)	−2.5	4.0	1.2	−7.2

Table 6

7.8 Table 7: Calendar-time portfolio regressions of recent flow-induced transactions

Read: Long recent fire-sale stocks and short recent inflow-purchase stocks produce economically meaningful alphas after factor controls.

Detailed interpretation: Event-study reversals translate into implementable portfolio evidence.

Economic message. Liquidity provision earns abnormal returns.

Table 7

Calendar-time portfolio regressions of flow-induced mutual fund transactions (1990-2004)

Dependent variables are event portfolio returns, R_{it} , in excess of the one-month Treasury bill rate, R_{t-1}^T , observed at the beginning of the month. Each month we form equal- and value-weighted portfolios of all sample firms that have completed the event within a one-year window. Lagged one quarter. The event portfolio is rebalanced quarterly to drop all stocks that reach the end of their event period and add all companies that have recently been involved in a flow-driven transaction. The three Fama-French factors are zero-investment portfolios representing the excess returns of the market, $R_m - R_{t-1}^T$, the difference between a portfolio of "small" stocks and "big" stocks, SMB, and the difference between a portfolio of "high" book-to-market stocks and "low" book-to-market stocks, HML. The fourth factor, UMD, is the difference between a portfolio of stocks with high past one-year returns minus a portfolio of stocks with low past one-year returns. Stocks recently experiencing outflow-induced sales are identified as those with an average $PRESSURE_{t-1}$ below the 10th percentile. Stocks recently experiencing inflow-induced purchases are identified as those with an average $PRESSURE_{t-1}$ above the 90th percentile. The firm average of $PRESSURE_{t-1}$ is calculated over the previous year, skipping the most recent quarter. A minimum of ten firms in the event portfolio is required. N denotes the number of monthly observations and t -statistics are in parentheses.

$$R_{P,t+1} - R_{t+1}^T = \alpha + \beta(R_{m,t+1} - R_{t+1}^T) + \beta_{SMB}(SMB_{t+1}) + \beta_{HML}(HML_{t+1}) + \beta_{UMD}(UMD_{t+1}) + \epsilon_{t+1}$$

Equally-weighted						Value-weighted					
Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N	Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N
<i>A. Recent outflow-induced sales</i>											
0.47%	0.0075				0.6762	0.29%	0.0091				0.6547
(2.11)	(19.28)				180	(1.49)	(18.37)				180
0.05%	1.1765	0.4123	0.6317		0.8078	-0.037%	0.0068	0.1489	0.4990		0.7557
(0.28)	(25.56)	(8.13)	(10.12)		180	(-0.16)	(22.87)	(3.13)	(8.51)		180
0.13%	1.1553	0.4221	0.6159	-0.0712	0.8122	-0.037%	0.0088	0.1484	0.4998	0.0038	0.7557
(0.72)	(24.58)	(8.36)	(9.87)	(-2.01)	180	(-0.18)	(22.14)	(3.09)	(8.44)	(0.11)	180

7.9 Table 8: Calendar-time portfolio regressions of expected flow-induced transactions

Read: Expected flows forecast future forced trading and generate tradable return patterns.

Detailed interpretation: This is the front-running result.

Economic message. Predictable flow pressure can destabilize prices before forced trades occur.

Table 8

Calendar-Time Portfolio Regressions of Anticipated Mutual Fund Forced Transactions (1990-2004)

Dependent variables are event portfolio returns, R_{it} , in excess of the one-month Treasury bill rate, R_{t-1}^T , observed at the beginning of the month. Each month we form equal- and value-weighted portfolios of all sample firms that are expected to complete the event within the next month. The three Fama-French factors are zero-investment portfolios representing the excess return of the market, $R_m - R_{t-1}^T$, the difference between a portfolio of "small" stocks and "big" stocks, SMB, and the difference between a portfolio of "high" book-to-market stocks and "low" book-to-market stocks, HML. The fourth factor, UMD, is the difference between a portfolio of stocks with high past one-year returns minus a portfolio of stocks with low past one-year returns. Anticipated fire sales are identified each month as stocks with $E[PRESSURE_{t+1}]$ in the bottom decile. Anticipated inflow-driven purchases are identified each month as stocks with $E[PRESSURE_{t+1}]$ in the top decile. $E[PRESSURE_{t+1}]$ is a stock-level variable, calculated each month as the number of shares held by mutual funds with expected flows above the 90th percentile, minus the number of shares held by mutual funds with flows below the 10th percentile, scaled by the average monthly volume of the stock, requiring at least ten owners. A minimum of ten firms in the event portfolio is required. The number of monthly observations is denoted by N and t -statistics are in parentheses.

$$R_{P,t+1} - R_{t+1}^T = \alpha + \beta(R_{m,t+1} - R_{t+1}^T) + \beta_{SMB}(SMB_{t+1}) + \beta_{HML}(HML_{t+1}) + \beta_{UMD}(UMD_{t+1}) + \epsilon_{t+1}$$

Equally-weighted						Value-weighted					
Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N	Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N
<i>A. Stocks expected to be in an inflow-driven purchase over the next month</i>											
0.34%	0.8930				0.6502	0.55%	0.8094				0.6722
(1.31)	(18.82)				156	(2.39)	(17.77)				156
0.39%	0.9536	0.5243	0.3322		0.8027	0.35%	0.9139	0.3405	0.1830		0.7391
(2.17)	(20.33)	(10.82)	(5.52)		156	(1.73)	(17.30)	(6.54)	(2.70)		156
0.29%	0.9848	0.5062	0.3484	0.0861	0.8109	0.08%	0.2905	0.2276	0.2385	0.8044	
(1.62)	(20.66)	(10.52)	(5.86)	(2.55)	156	(0.45)	(21.07)	(6.06)	(3.84)	(7.10)	156
<i>B. Stocks expected to be in a fire sale over the next month</i>											
-0.20%	1.0011				0.6277	-0.14%	0.9017				0.5967
(-0.74)	(18.11)				156	(-0.44)	(15.09)				156
-0.68%	1.1763	0.5009	0.5230		0.7182	-0.51%	1.1081	-0.0599	0.4355		0.6059
(-2.82)	(18.39)	(5.46)	(6.37)		156	(-2.54)	(17.76)	(-0.79)	(5.44)		156
-0.36%	1.0711	0.4216	0.4687	-0.2899	0.7992	-0.27%	1.8288	-0.0052	0.3946	-0.2184	0.7331
(-1.68)	(18.66)	(7.28)	(6.54)	(-1.14)	156	(-1.19)	(17.24)	(-0.09)	(5.30)	(-5.17)	156

7.10 Table 9: Return predictability of anticipated pressure

Read: Anticipated pressure predicts cross-sectional returns in several specifications.

Detailed interpretation: The pressure mechanism becomes part of the return-predictability structure of equity markets.

Economic message. Predictable institutional demand shocks create incentives for anticipatory trading.

B. Recent inflow-induced purchases

-0.50%	1.3028				0.7775	-0.49%	1.1059				0.7846
(-2.20)	(24.94)				180	(-2.58)	(25.47)				180
-0.04%	1.3444	0.2681	0.2129		0.7989	-0.62%	1.1883	0.0243	0.2107		0.7972
(-2.84)	(23.20)	(4.20)	(2.71)		180	(-3.25)	(24.17)	(0.45)	(3.16)		180
-0.32%	1.2487	0.3087	0.1476	-0.2657	0.8476	-0.44%	1.1569	0.0461	0.1757	-0.1576	0.8169
(-1.81)	(23.93)	(5.52)	(2.18)	(-7.48)	180	(-2.35)	(23.52)	(0.89)	(2.76)	(-4.33)	180
<i>C. Long recent outflow-induced sales and short recent inflow-induced purchases</i>											
0.97%	-0.3153				0.1556	0.77%	-0.2968				0.1555
(3.98)	(-5.64)				180	(3.40)	(-5.58)				180
0.71%	-0.1679	0.1442	0.4188		0.2596	0.59%	-0.1997	0.1246	0.2883		0.2128
(3.01)	(-2.76)	(2.15)	(5.07)		180	(2.60)	(-3.40)	(1.52)	(3.62)		180
0.45%	-0.0954	0.1135	0.4662	0.2225	0.3528	0.41%	-0.1471	0.1023	0.3241	0.1613	0.2687
(2.00)	(-1.82)	(1.80)	(6.00)	(5.02)	180	(1.81)	(-2.51)	(1.63)	(4.17)	(3.66)	180

Table 8 (continued)

Equally-weighted						Value-weighted					
Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N	Intercept	Rm - R ^T	SMB	HML	UMD	R ² /N
<i>C. Long stocks expected to be in an inflow-driven purchase over the next month and short stocks expected to be in a fire sale over the next month</i>											
0.94%	-0.1081				0.0155	0.71%	-0.0125				0.0001
(3.18)	(-1.56)				156	(1.86)	(-0.14)				156
1.07%	-0.2228	0.1634	-0.1907		0.0988	0.86%	-0.1942	0.3914	-0.2525		0.2041
(3.59)	(-2.84)	(2.01)	(-1.89)		156	(2.50)	(-2.15)	(4.30)	(-2.18)		156
0.65%	-0.0863	0.0646	-0.1203	0.3760	0.3524	0.35%	-0.0281	0.2957	-0.1670	0.4569	0.4553
(2.51)	(-1.25)	(1.21)	(-1.40)	(7.69)	156	(1.20)	(-0.37)	(3.79)	(-1.73)	(8.34)	156

Table 9
Fama-macbeth cross-sectional regressions of monthly stock returns (1990–2004).
Dependent variables are monthly stock returns, $R_{i,t+1}$, in excess of the one-month Treasury bill rate, R_t^f , observed at the beginning of the month. $E[PRESSURE_{i,t+1}]$ is a stock-level variable, calculated each month as the number of shares held by mutual funds with expected flows above the 90th percentile, minus the number of shares held by mutual funds with flows below the 10th percentile, scaled by the average monthly volume of the stock, requiring at least ten owners. *Average Pressure* is the one-year average value of *PRESSURE*, lagged one-quarter. Beta is the slope coefficient from a 36-month market model regression, using the CRSP value-weighted portfolio as the market proxy. BE/ME is the ratio of book equity to market equity following the definition in Fama-French (1993). R_t is the one-month lagged return; $R_{t-12,t-1}$ is the 11-month lagged return ending one month prior; $R_{t-36,t-13}$ is the 24-month lagged return ending 12 months prior. Net Equity Issuance measures the three-year composite share issuance variable defined by Daniel and Titman (2006). The table reports average coefficients from 151 monthly cross-sectional regressions. The averages are time-series means with *t*-statistics (in parentheses) corresponding to the standard error of the mean. R^2 is the average cross-sectional R^2 . N denotes the average number of cross-sectional observations.

Int	$E[PRESSURE_{i,t+1}]$	Average pressure _{t-3}	Beta	Size	BE/ME	R_t	$R_{t-12,t-1}$	$R_{t-36,t-13}$	Net equity issuance	R^2 [N]
<i>PRESSURE_1</i>										
0.0096	0.0096	−0.0257								0.0085
(2.30)	(2.30)	(−2.34)								[1.017]
0.0178	0.0026	−0.0171	0.0022	−0.0006	0.0010	−0.0278	0.0086	−0.0019	−0.0042	0.1060
(1.36)	(0.98)	(−2.46)	(0.86)	(−0.79)	(1.05)	(−3.50)	(2.50)	(−1.16)	(−3.54)	[1.017]
<i>PRESSURE_2</i>										
0.0095	0.1281	−0.1595								0.0149
(2.51)	(2.00)	(−2.89)								[1.017]
0.0159	0.0689	−0.0991	0.0022	−0.0005	0.0010	−0.0281	0.0086	−0.0018	−0.0044	0.1074
(1.23)	(1.93)	(−2.60)	(0.86)	(−0.66)	(1.08)	(−3.55)	(2.55)	(−1.10)	(−3.69)	[1.017]
<i>PRESSURE_3</i>										
0.0095	0.1232	−0.0004								0.0186
(2.51)	(1.98)	(−2.44)								[1.017]
0.0157	0.0655	−0.0002	0.0024	−0.0005	0.0010	−0.0282	0.0086	−0.0015	−0.0043	0.1075
(1.22)	(1.84)	(−2.23)	(0.93)	(−0.67)	(1.11)	(−3.56)	(2.52)	(−0.95)	(−3.54)	[1.017]

Table 9

8 Detailed result map

- Fund flows are predictable.
- Flows change fund trading behavior.
- Flow-pressure stocks are not only small illiquid stocks.
- Forced selling generates temporary negative price pressure and reversal.
- Voluntary sales provide a placebo comparison.
- Alternative pressure measures confirm robustness.
- Funds cannot fully avoid pressure costs through trade selection.
- Liquidity provision and anticipated-pressure strategies earn abnormal returns.

9 Short summary

- The paper studies institutional price pressure using mutual fund flows and holdings.
- Funds with large outflows sell existing positions; funds with large inflows buy more of existing positions.
- Stocks held by many distressed funds experience fire-sale pressure.
- Fire-sale stocks decline and then rebound, consistent with temporary mispricing.
- Calendar-time strategies trading against pressure earn abnormal returns.
- Predictable fund flows create incentives for front-running future forced trades.

10 Conclusion

10.1 Core conclusion

Coval and Stafford show that asset fire sales occur in listed equity markets. The key is clustered institutional pressure from overlapping mutual fund holdings.

10.2 Incremental contribution

The paper provides a transparent empirical setting for measuring fire-sale costs, documents both forced sales and inflow-driven purchases, and connects mutual fund organizational design to temporary price pressure.

10.3 Economic intuition

A stock can become temporarily illiquid when many similar institutions need to trade in the same direction. Prices move enough to attract outside liquidity and then reverse when pressure dissipates.

10.4 Implications

The findings imply that short-run demand curves for stocks are less than perfectly elastic and that rational mispricing can arise without irrational investors.

10.5 Limitations

Quarterly holdings limit exact trade timing, and some trades may combine information and liquidity motives. Still, the reversal evidence makes a pure information story unlikely.

10.6 Takeaway

Liquidity is endogenous. Market organization and investor flows can create temporary mispricing even in highly liquid equity markets.